Design and Implementation of a Microstrip Patch Antenna for ISM Band Applications

1. **Abstract:** In this work, a microstrip patch antenna is designed, simulated, and fabricated for ISM band applications. The antenna is designed using CST Microwave Studio and optimized by analyzing the effects of various geometrical parameters on its performance. The simulated results show that the antenna resonates at 2.44 GHz with a return loss of -13.81 dB, but exhibits low radiation efficiency of 30% and total efficiency of 8%, with a realized gain of -10 dBi at the resonant frequency. The fabricated prototype is measured using a Vector Network Analyzer (VNA), and the measured results demonstrate a strong resonance at 3.95 GHz with an S11 of -32.37 dB. A significant frequency shift of approximately 1.51 GHz is observed between the simulated and measured results, which may be attributed to discrepancies in substrate properties, fabrication tolerances, and simulation setup inaccuracies. The low simulated efficiency and negative gain indicate that further optimization of the antenna geometry and feeding mechanism is required to achieve satisfactory performance for ISM band applications.

2. Objectives

I. To design and optimize a microstrip patch antenna for ISM band (2.45 GHz) applications using CST Microwave Studio by analyzing the effects of various geometrical parameters on antenna performance.

II. To fabricate the optimized antenna prototype and measure its S_{11} characteristics using a Vector Network Analyzer (VNA).

III. To compare the simulated and measured results and analyze the discrepancies to validate the antenna design.

IV. To evaluate the gain, efficiency, and radiation performance of the proposed antenna for practical ISM band applications.

3. Antenna Design

The proposed antenna is designed using CST Microwave Studio on an FR-4 lossy substrate with a dielectric constant (εr) of 4.3 and a thickness of 1.6 mm. The top layer consists of a microstrip patch made of annealed copper, while the bottom layer serves as the ground plane, also made of annealed copper. The antenna is fed using a microstrip transmission line to achieve proper impedance matching at the desired operating frequency

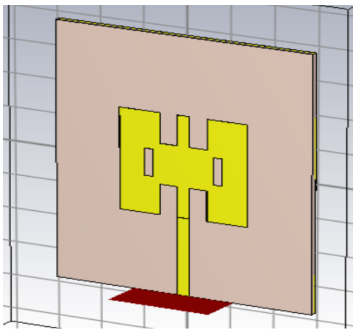


Fig.1: Geometry of the proposed antenna.

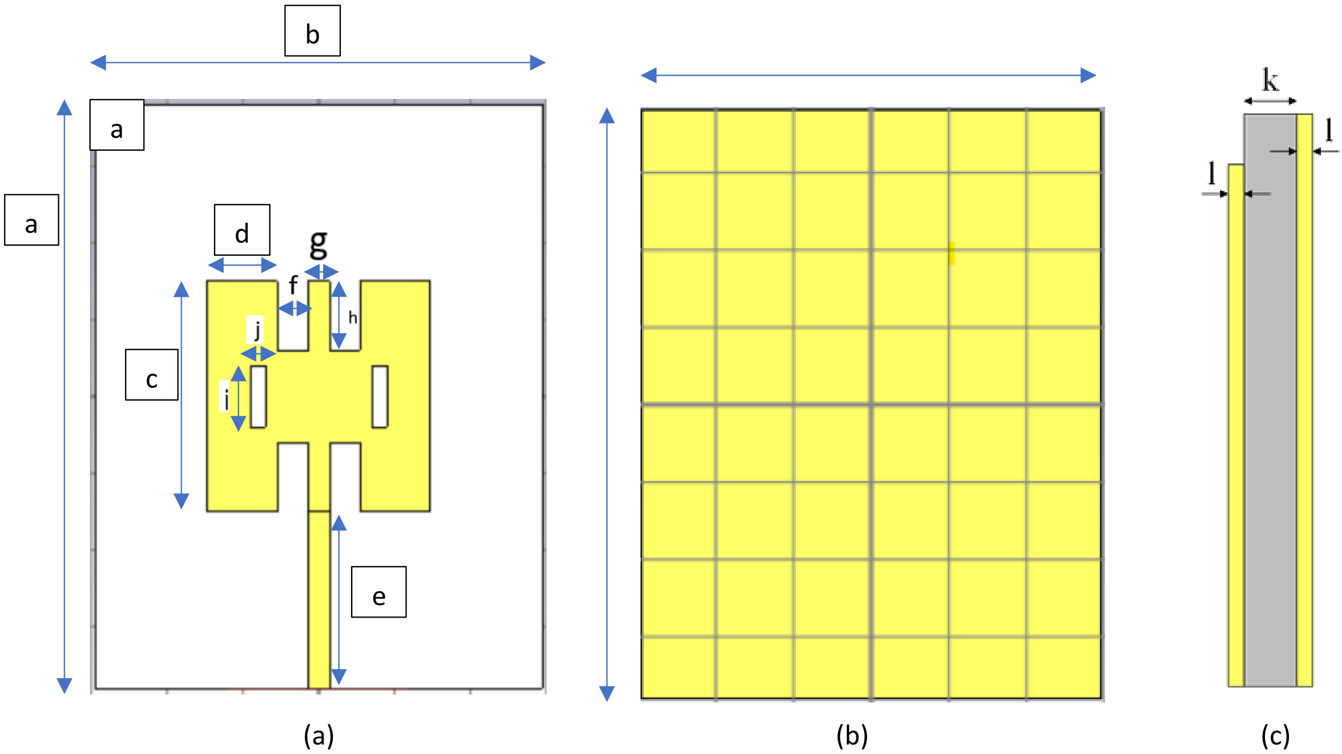
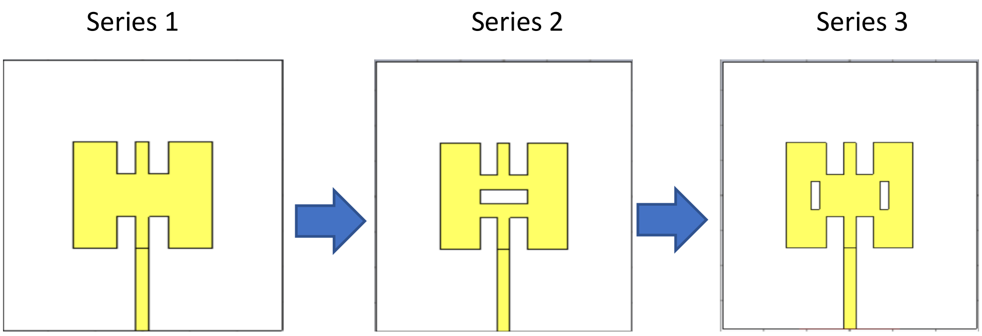


Fig.2: (a) Top view, (b) bottom view and (c) side view of the proposed antenna.

Table 1: Geometrical dimensions.

Parameter	Size (mm)
a	76
b	59
c	30
d	10.75
e	23
f	4
g	2.86
h	9
i	9
j	2
k	1.5
l	0.035

4. Effect of Geometrical Variations on the Antenna Performance



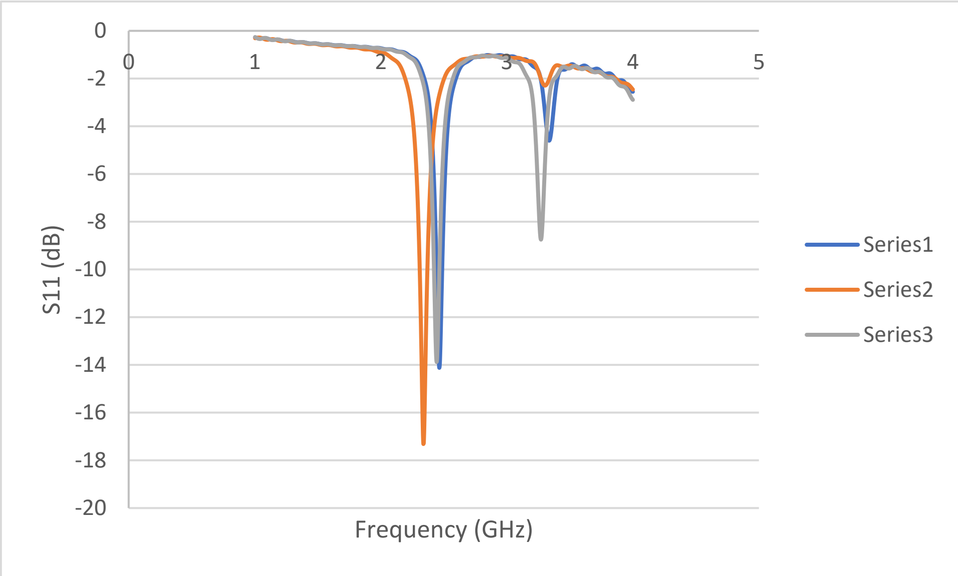


Fig.3: Evolution process of the proposed antenna.

5. Results and Discussions

5.1. Simulated Results

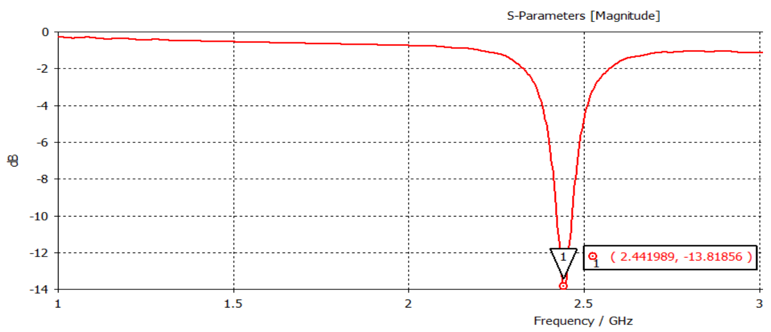


Fig.5: Simulated reflection coefficients (S11) of the proposed antenna.

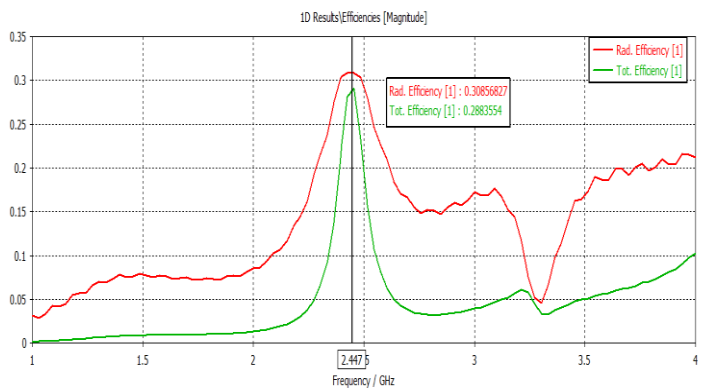


Fig.6: Efficiency of the proposed antenna.

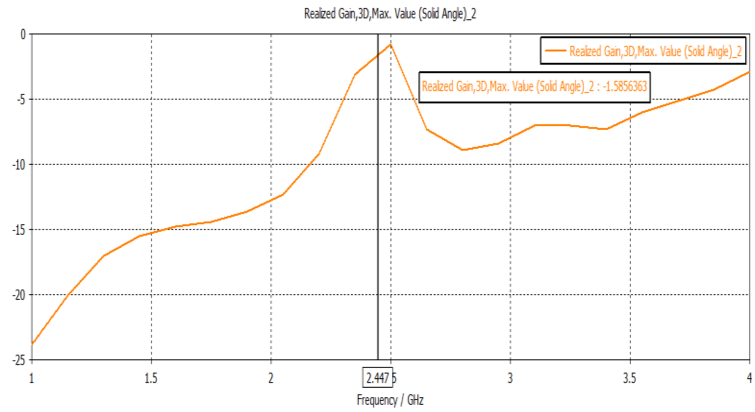


Fig.7: Variation of Gain w.r.t. frequency of the proposed antenna.

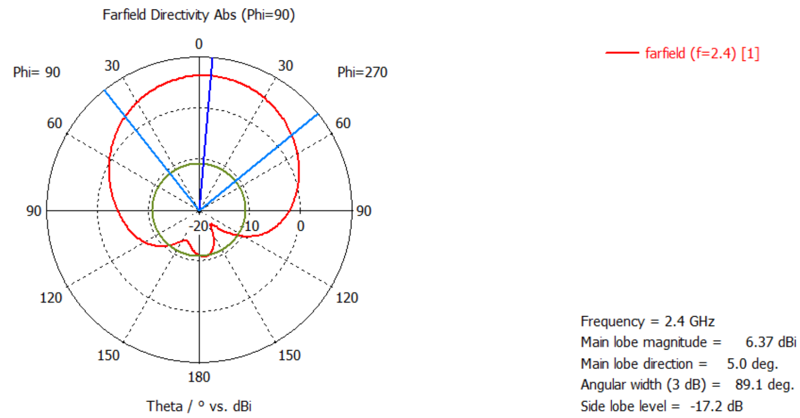


Fig.8: 1D radiation pattern at 2.44 GHz in E-plane

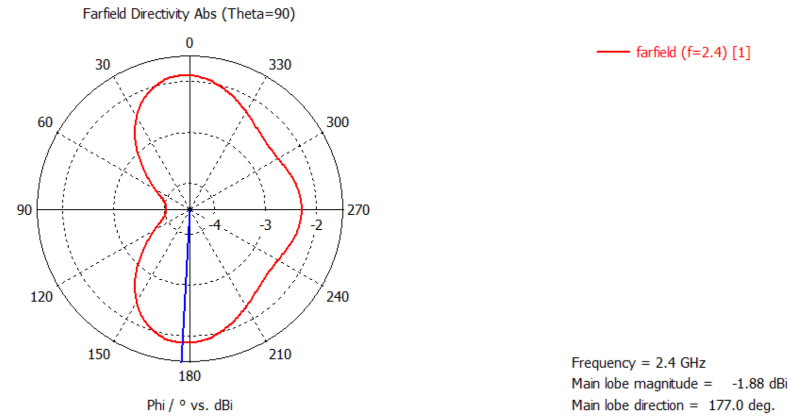


Fig.9: 1D radiation pattern at 2.44 GHz in H-plane

5.2. Experimental Results

Fig.10: (a) Front view and (b) Back view of the fabricated antenna and (c) experimental setup.

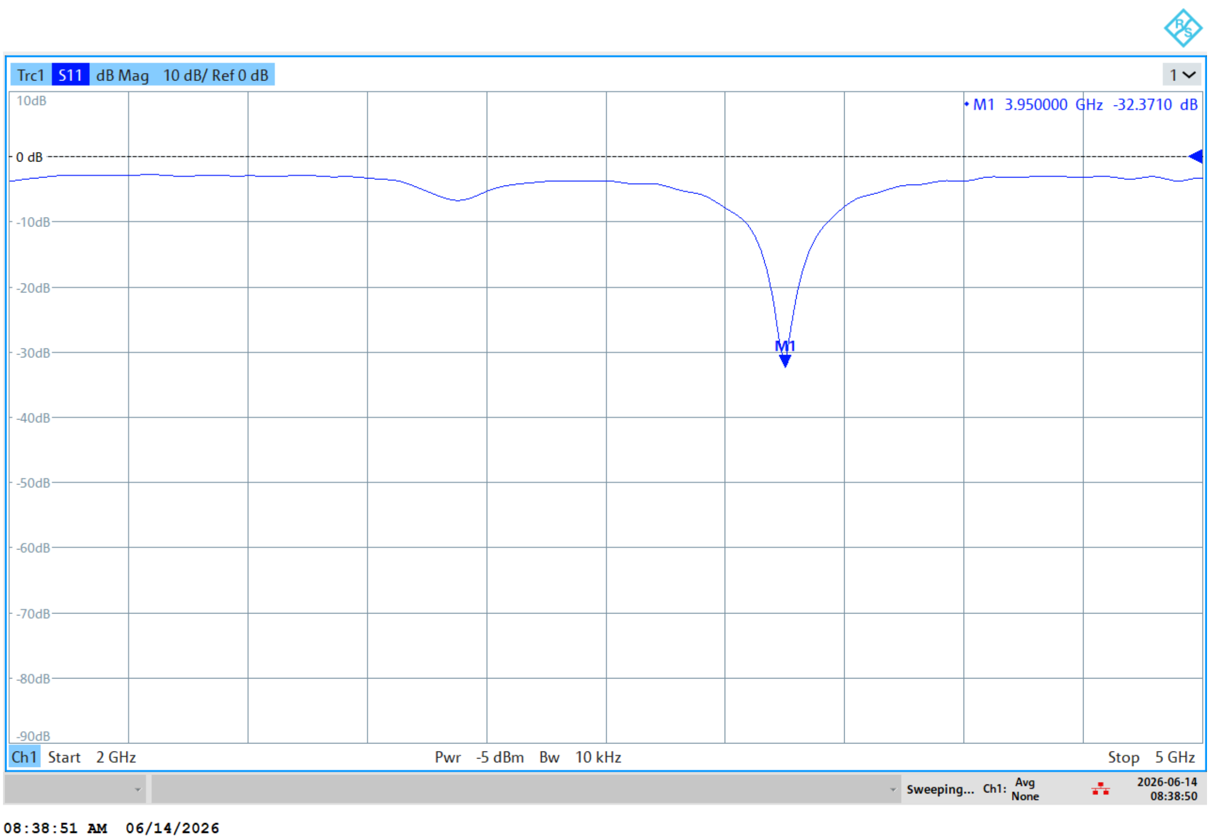
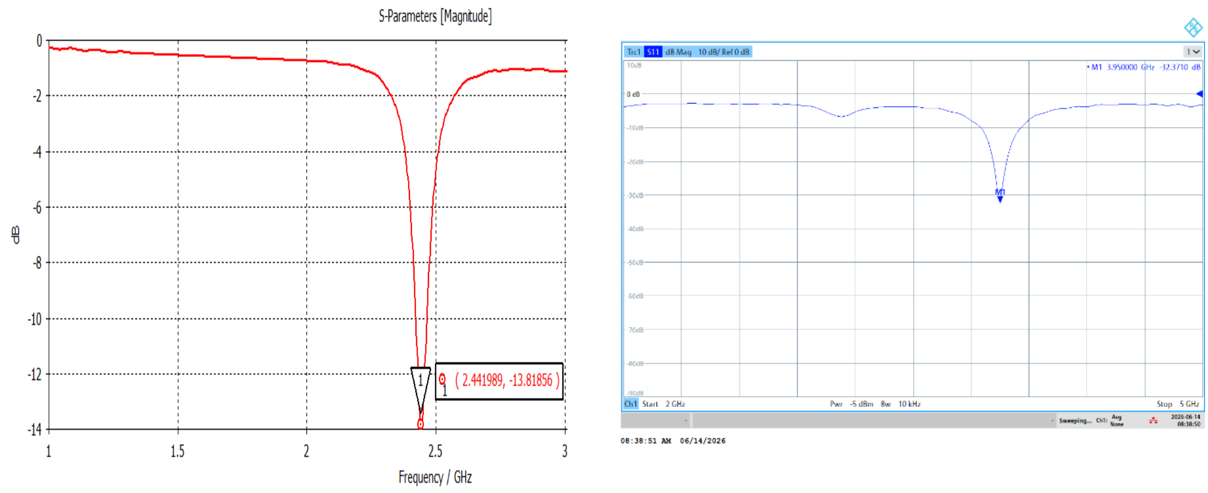


Fig.11: Experimental results of the antenna.

6. Comparison between Simulated and Measured Results



(a) **(b)**

Fig.12: Comparison between simulated and measured results (a) Simulated. (b) Measured.

7. Addressing Complex Engineering Problems (P's)

P1 (Depth of Knowledge): This project applies microwave engineering principles including transmission line theory, impedance matching, and microstrip antenna design. CST Microwave Studio is used for simulation, requiring knowledge of numerical techniques and S-parameter analysis. Fabrication and VNA measurements demand understanding of substrate properties, calibration techniques, and radiation performance evaluation.

P2 (Range of Conflicting Requirements): The design involves trade-offs between return loss and bandwidth, gain and antenna size, efficiency and fabrication complexity. Additionally, simulated performance often conflicts with measured results due to ideal assumptions versus real-world imperfections such as fabrication tolerances and connector effects. These conflicts require careful optimization to achieve a practical solution.

P4 (Familiarity of Issues): The project addresses practical issues including fabrication tolerances, substrate parameter variations, SMA connector parasitic effects, VNA measurement uncertainties, EMI concerns, and cost/manufacturing constraints. These reflect real-world challenges encountered in microwave system development.

8. Conclusion

In this project, a microstrip patch antenna was successfully designed, simulated, and fabricated for ISM band applications. The antenna was modeled using CST Microwave Studio, and the effects of various geometrical parameters on its performance were analyzed. The simulated results showed resonance at 2.44 GHz with a return loss of -13.81 dB, while the measured results exhibited resonance at 3.95 GHz with a return loss of -32.37 dB. A significant frequency shift of 1.51 GHz was observed between the simulated and measured results, which may be attributed to fabrication tolerances, substrate parameter variations, and SMA connector effects.